

EDUCATION

Michigan State University, East Lansing, MI

*Bachelor of Science, **Electrical Engineering**, (Minor in Mathematics)*

Affiliations: IEEE MSU Student Branch (**Technical Director**)

Honors: 2nd Place, MSU Research Forum 2026 · \$100K merit scholarship · Professorial Assistantship

GPA: 4.00

Expected May 2029

EXPERIENCE

MSU Solar Car Racing

East Lansing, MI

Harnessing and Integration Lead

September 2025 – Present

- Developed and deployed STM32 firmware in C++ for an on-vehicle controller, implementing real-time CAN and LIN communication for 5 vehicle signals and faults, including turn signals, hazards, braking, and battery protection faults
- Engineered safety-critical vehicle control electronics from schematic capture through impedance-controlled PCB layout in Altium Designer, validating MCUs, sensors, MOSFETs, and regulators against operating constraints
- Designed and built a multi-protocol bus analyzer on an STM32C552 that monitors and transmits on CAN-FD, LIN, SPI, UART, and I2C, streaming decoded traffic to a host-side GUI used as the team's debug tool
- Traced recurring shorts to high-voltage isolation faults in the vehicle harness; stripped and rebuilt wiring for the control, battery management, motor, and steering-wheel systems in 5 days, restoring safe drivable operation

RF-Photonics Lab, Michigan State University

East Lansing, MI

Undergraduate Researcher

June 2026 – Present

- Built 50+ GNU Radio flowgraphs and native UHD Python transmit/receive software implementing FIR filtering, pulse shaping, symbol-timing recovery, and QPSK/16-QAM/256-QAM modulation on the USRP B210
- Cut square-wave rise/fall time from 135 ns to 45 ns on the USRP B210 by tuning sample rate, frequency, and gain, characterizing link quality with BER and EVM and resolving underflow and cable-length induced phase offsets
- Transmitted OFDM data and an LFM chirp concurrently in the same frequency band at different bandwidths, demonstrating simultaneous communication and radar sensing on shared spectrum

Microtechnology Lab, Michigan State University

East Lansing, MI

Undergraduate Researcher

September 2025 – Present

- Developed bare-metal C/C++ firmware on a Teensy 4.1 implementing real-time ADC/DAC control loops and timing-critical waveform generation, with Python and MATLAB tooling for system operation and data analysis
- Conducted 1,000+ electrochemical experiments detecting heavy metal ions down to 10 ppb using cyclic voltammetry, differential pulse voltammetry, and fast-scan CV with carbon fiber and boron-doped diamond working electrodes
- Built a 3D COMSOL model of a Tesla-valve microfluidic channel, solving laminar flow and species concentration to determine electrode placement, validated against measured sensor response

PROJECTS

- **ANC Engine:** Engineered a fixed-point FxLMS active noise cancellation engine that converges to -34 dB noise reduction in 50 ms certified against a self-built deterministic acoustic simulator and 12-case robustness suite
- **RISC-V CPU & SoC:** Designed a complete RV32I processor and SoC in Verilog that passes all 41 official RISC-V instruction tests and runs C compiled with stock GCC synthesized to a Gowin FPGA at 27 MHz
- **Automotive ECU Platform:** Built 4-node CAN-FD vehicle network on STM32 in C++ implementing embedded security via a secure bootloader supporting A/B image slots, ECDSA-P256 signature verification, and watchdog-based rollback, reflashing nodes over bus through ISO-TP (ISO 15765-2) transport and UDS (ISO 14229) diagnostic server
- **Game Boy Emulator:** Built a cycle-accurate Game Boy emulator in C++ passing 34/34 Blargg and 91/93 Mooneye timing tests, and rendered pixel-perfect dmg-acid2 output
- **NAND Flash Characterization:** Built microcontroller-driven rig characterizing SPI NAND endurance and retention, logging BER vs. P/E cycles and temperature-accelerated bit decay, analyzed with Weibull failure-distribution fits

SKILLS

- **Hardware Design:** Altium Designer, KiCad, LTspice, AutoCAD, SolidWorks, schematic capture, PCB layout
- **Programming & Analysis:** C, C++, Assembly, Python, MATLAB, Origin, COMSOL Multiphysics, Git, Ansys
- **Automotive & Protocols:** CAN, CAN-FD, LIN, ISO-TP (ISO 15765-2), UDS (ISO 14229), I2C, SPI, Secure Boot
- **Digital Design:** Verilog, RTL design, RISC-V ISA, FPGA synthesis, Icarus Verilog
- **Signal Processing & SDR:** GNU Radio, QPSK, QAM, OFDM, FMCW, LFM, BER and EVM characterization
- **Lab Equipment & Testing:** Oscilloscope, Multimeter, Signal generator, Spectrum analyzer, Logic analyzer